ADHD Focus Coach

API Documentation — Version 1.0

Base URL: `/api/v1`

Authentication: Protected endpoints use `Authorization: Bearer <Clerk JWT>`. FastAPI verifies the token and resolves the application user.

Architecture: Mobile → FastAPI → Services → PostgreSQL / Redis / AI. PostgreSQL is the persistent source of truth; Redis is used for cache and rate limiting; AI generates/adapts content.

1. Authentication

Current User

GET	`/auth/me`
Purpose	Returns the authenticated application user.

Success response

```
{"id":12,"clerk_user_id":"user_xxx","role":"CAREGIVER","created_at":"2026-09-23T10:00:00Z"}
```

Errors

Code	Meaning
401	Invalid or missing authentication

2. Children

Create Child

POST	`/children`
Purpose	Creates a child profile for the authenticated caregiver.

Request

```
{"name":"Aarav","age":10,"class_name":"5","description":"Prefers short visual instructions."}
```

Success response

```
{"id":1,"caregiver_id":12,"name":"Aarav","age":10,"class_name":"5","description":"Prefers short visual instructions."}
```

Errors

Code	Meaning
400	Invalid data
401	Unauthorized
403	User is not a caregiver

List Children

GET	`/children`
Purpose	Returns children belonging to the authenticated caregiver.

Success response

```
[{"id":1,"name":"Aarav","age":10,"class_name":"5","description":"Prefers short visual instructions."}]
```

Errors

Code	Meaning
401	Unauthorized

Get Child

GET	`/children/{child_id}`
Purpose	Returns a specific child profile.

Errors

Code	Meaning
401	Unauthorized
403	Not authorized
404	Child not found

Update Child

PATCH	`/children/{child_id}`
Purpose	Updates editable child profile information.

Request

```
{"age":11,"description":"Responds well to encouragement."}
```

Success response

```
Returns the updated child profile.
```

Errors

Code	Meaning
401	Unauthorized
403	Not authorized
404	Child not found
422	Validation failure

3. Tasks

Create Task

POST	`/tasks`
Purpose	Creates a task for an authorized child.

Request

```
{"child_id":1,"title":"Math Homework","description":"Complete questions 1-5","scheduled_at":"2026-09-23T18:00:00Z","priority":1}
```

Success response

```
{"id":42,"child_id":1,"title":"Math Homework","status":"PENDING","focus_state":"INITIATED","current_step_position":null,"last_activity_at":null}
```

Errors

Code	Meaning
400	Invalid data
401	Unauthorized
403	Child is not accessible

List Tasks

GET	`/tasks`
Purpose	Returns tasks accessible to the authenticated user.

Errors

Code	Meaning
401	Unauthorized

Query: `child_id` and `status` are optional.

Get Task

GET	`/tasks/{task_id}`
Purpose	Returns a task and its generated steps.

Success response

```
{"id":42,"child_id":1,"title":"Math Homework","status":"PENDING","focus_state":"INITIATED","current_step_position":null,"steps":[{"id":101,"position":1,"description":"Open your mathematics notebook","estimated_minutes":1,"status":"PENDING"}]}
```

Errors

Code	Meaning
401	Unauthorized
403	Not authorized
404	Task not found

Update Task

PATCH	`/tasks/{task_id}`
Purpose	Updates editable task information.

Request

```
{"title":"Math Homework","scheduled_at":"2026-09-23T19:00:00Z","priority":2}
```

Success response

```
Returns the updated task.
```

Errors

Code	Meaning
401	Unauthorized

403	Not authorized
404	Task not found
422	Validation failure

Delete / Cancel Task

DELETE	`/tasks/{task_id}`
Purpose	Deletes or cancels a task according to its current state.

Success response

```
204 No Content
```

Errors

Code	Meaning
401	Unauthorized
403	Not authorized
404	Task not found
409	Task cannot be cancelled in its current state

4. AI Task Breakdown

Generate Breakdown

POST	`/tasks/{task_id}/breakdown`
Purpose	Generates a structured 3–5 step breakdown using the configured AI provider.

Success response

```
{"task_id":42,"steps":[{"description":"Open your mathematics notebook","estimated_minutes":1},{"description":"Write today's date","estimated_minutes":1},{"description":"Solve question 1","estimated_minutes":3}]}
```

Errors

Code	Meaning
401	Unauthorized
403	Task inaccessible
404	Task not found
422	AI output failed validation
429	Rate limit exceeded
503	AI provider unavailable

Validation: 3–5 steps; each <5 minutes; Pydantic/structured validation; business validation; safety validation. Invalid output is rejected/regenerated.

5. Focus Execution

Start Focus

POST	`/tasks/{task_id}/start`
Purpose	Starts focus execution and initializes task FSM/runtime state.

Success response

```
{"task_id":42,"focus_state":"ACTIVE","current_step_position":1,"last_activity_at":"2026-09-23T18:00:00Z"}
```

Errors

Code	Meaning
401	Unauthorized
403	Not authorized
404	Task not found
409	Task cannot be started

Get Focus State

GET	`/tasks/{task_id}/focus-state`
Purpose	Returns current focus state and active step.

Success response

```
{"task_id":42,"focus_state":"ACTIVE","current_step_position":2,"last_activity_at":"2026-09-23T18:04:20Z"}
```

Errors

Code	Meaning
401	Unauthorized
403	Not authorized

404	Task not found

Complete Step

POST	`/tasks/{task_id}/steps/{step_id}/complete`
Purpose	Completes the current step and determines the next executable step.

Success response


```
{"task_id":42,"completed_step_id":101,"next_step_position":2,"task_status":"IN_PROGRESS","focus_state":"ACTIVE"}
```


Errors

Code	Meaning
401	Unauthorized
403	Not authorized
404	Task/step not found
409	Step is not the current executable step

Report Stuck

POST	`/tasks/{task_id}/steps/{step_id}/stuck`
Purpose	Reports that the child is stuck and triggers deterministic adaptation.

Request


```
{"reason":"TOO_DIFFICULT"}
```


Success response


```
{"task_id":42,"step_id":103,"action":"DECOMPOSE","steps":[{"description":"Read the first part of the question","estimated_minutes":2},{"description":"Identify the values given","estimated_minutes":2},{"description":"Write the first calculation","estimated_minutes":3}]}
```


Errors

Code	Meaning
400	Invalid stuck reason
401	Unauthorized
403	Not authorized
404	Task/step not found
422	AI adaptation failed validation
429	Rate limit exceeded
503	AI provider unavailable

Reasons: DONT_UNDERSTAND, DONT_KNOW_WHERE_TO_START, TOO_DIFFICULT, STEP_TOO_BIG, NEED_A_BREAK. Understand/start → REFINE; difficult/big → DECOMPOSE; break → BREAK.

6. Feedback

Submit Feedback

POST	`/feedback`
Purpose	Stores the child's post-task FAQ/feedback.

Request


```
{"task_id":42,"task_step_id":105,"feeling":"A_LITTLE_HARD","helped_by":"SMALL_STEPS","difficulty":"TOO_LONG"}
```


Success response


```
{"id":15,"child_id":1,"task_id":42,"task_step_id":105,"feeling":"A_LITTLE_HARD","helped_by":"SMALL_STEPS","difficulty":"TOO_LONG","created_at":"2026-09-23T18:30:00Z"}
```


Errors

Code	Meaning
401	Unauthorized
403	Not authorized
404	Task/step not found
422	Validation failure

7. Progress

Get Progress

GET	`/progress`
Purpose	Returns progress data for the authorized child context.

Success response

{"tasks_completed":18,"tasks_total":24,"completion_rate":75,"current_streak":4}

Errors

Code	Meaning
401	Unauthorized
403	Not authorized
404	Child not found

Progress Summary

GET	`/progress/summary`
Purpose	Returns summarized task and focus metrics.

Success response

{"completed_tasks":18,"steps_completed":62,"stuck_events":9,"inactivity_events":6}

Errors

Code	Meaning
401	Unauthorized
403	Not authorized

Get Streak

GET	`/progress/streak`
Purpose	Returns current and longest completion streak.

Success response

{"current_streak":4,"longest_streak":7}

Errors

Code	Meaning
401	Unauthorized
403	Not authorized

8. Notifications

List Notifications

GET	`/notifications`
Purpose	Returns notifications available to the authenticated user.

Success response

[{"id":8,"child_id":1,"task_id":42,"type":"TASK_REMINDER","message":"Math homework starts in 10 minutes.","scheduled_at":"2026-09-23T17:50:00Z","status":"PENDING"}]

Errors

Code	Meaning
401	Unauthorized

Query: `status` is optional. NotificationService controls status updates; no client PATCH endpoint is exposed.

9. Telemetry

Record Event

POST	`/telemetry/events`
Purpose	Records a meaningful product event for analytics and evaluation.

Request

{"task_id":42,"task_step_id":103,"event_type":"STEP_STUCK","metadata":{"reason":"TOO_DIFFICULT"}}

Success response

{"id":501,"child_id":1,"task_id":42,"task_step_id":103,"event_type":"STEP_STUCK","metadata":{"reason":"TOO_DIFFICULT"},"created_at":"2026-09-23T18:12:00Z"}

Errors

Code	Meaning
400	Invalid event
401	Unauthorized
403	Not authorized
404	Task/step not found
422	Validation failure

Events: TASK_STARTED, STEP_COMPLETED, STEP_STUCK, INACTIVITY_TRIGGERED, BREAK_STARTED, BREAK_COMPLETED, TASK_COMPLETED, FEEDBACK_SUBMITTED.

10. Common API Errors

Code	Meaning
400	Invalid request
401	Authentication required or invalid
403	Authenticated but not authorized
404	Resource not found
409	State or resource conflict
422	Validation failure
429	Rate limit exceeded
500	Internal server error
503	External service unavailable

11. Endpoint Summary

Module	Method	Endpoint
Auth	GET	/auth/me
Children	POST	/children
Children	GET	/children
Children	GET	/children/{child_id}
Children	PATCH	/children/{child_id}
Tasks	POST	/tasks
Tasks	GET	/tasks
Tasks	GET	/tasks/{task_id}
Tasks	PATCH	/tasks/{task_id}
Tasks	DELETE	/tasks/{task_id}
AI	POST	/tasks/{task_id}/breakdown
Focus	POST	/tasks/{task_id}/start
Focus	GET	/tasks/{task_id}/focus-state
Focus	POST	/tasks/{task_id}/steps/{step_id}/complete
Focus	POST	/tasks/{task_id}/steps/{step_id}/stuck
Feedback	POST	/feedback
Progress	GET	/progress
Progress	GET	/progress/summary
Progress	GET	/progress/streak
Notifications	GET	/notifications
Telemetry	POST	/telemetry/events